PROJECT AGERE • RESEARCH DOSSIER • STATISTICAL EVALUATION OF ENTIRE RESEARCH

Statistical & Mathematical Evaluation of Entire Research

Complete Cross-Tier Empirical Evidence, Paired Hypothesis Testing, Stage Funnel Decompositions & Formal Selective-Answering Framework

EVALUATED OUTPUT ROWS	UNIQUE FINQA QUESTIONS	DEPLOYMENT CEILINGS	TESTING PROTOCOLS
2,078 Total (1,728 FinQA + 350 Synthetic)	**261 Questions (80 Reports, 60+ Co.)**	**8 GB (6.40 GB Cap) • 16 GB (11.50 GB Cap)**	**Exact McNemar, Bowker, Stuart-Maxwell, Bootstrap**

EXECUTIVE STATISTICAL VERDICT & CORE EMPIRICAL FINDINGS

1. Single-Agent System (SAS) Architectures Deliver Dominant Useful Answer Accuracy: Across both evaluated workstation tiers, allocating the memory ceiling to a single larger quantized generalist model consistently yields statistically and practically superior useful answer correctness compared to resident full-precision specialist teams (MAS). At the primary 16 GB evaluation (N = 160 questions, 50 reports, 36 companies), 14B Q4 SAS delivers **78/160 (48.75%)** reported correct answers versus **5/160 (3.13%)** for resident 1.5B/3B/0.5B F16 MAS ($\hat{\Delta}$ = −45.625 pp, Exact McNemar $p = 2.12\times10^{-22}$, Holm-adjusted $p = 6.10\times10^{-5}$). Under raw execution, SAS achieves 38 correct versus MAS 5 ($\hat{\Delta}$ = −20.625 pp, $p = 2.10\times10^{-9}$). At the 8 GB Natural Selection benchmark (N = 52), SAS delivers 8–10 correct answers (15.4%–19.2%) versus MAS 3–4 (5.8%–7.7%), while at the 8 GB Reserved Diagnostic Audit (N = 49), SAS delivers 8–9 correct answers (16.3%–18.4%) versus MAS 4 (8.2%).

2. MAS Performance Bottleneck Is Candidate Generation, Not Verification: The extreme withholding of MAS at 16 GB (152/160 cases, 95.00%) and 8 GB (57.1%–73.5%) is overwhelmingly driven by initial candidate generation failure rather than verifier over-rejection. At 16 GB, MAS produces only **10/160 (6.25%)** executable program candidates (with 150 cases terminating with `INSUFFICIENT` decisions directly from the drafter). All 5 correct candidates are delivered (acceptance rate a = 1.000). By the Fixed-Candidate Gating Bound (Proposition 2), even a mathematically perfect review gate could deliver at most 5/160 correct answers from those candidate sets.

3. Novel Algorithmic Candidates Failed to Achieve Statistically Significant Improvements: None of the newly proposed algorithmic mechanisms achieved statistically significant QA gains: (i) *Crossed-question counterfactual curriculum* reduced wrong deliveries at 8 GB by shifting them to withheld, but failed the predeclared Pareto/no-correct-loss rule (losing 1 correct answer versus Strong Original Critic, Δ = −1.92%, p = 1.0000) and failed to replicate on the reserved partition (p = 1.0000); (ii) *Multi-stage iterative repair* achieved exact zero accuracy gain across all natural 8 GB and 16 GB runs (Δ_{corr} = 0.00%, p = 1.0000), while adding 16%–64% token overhead and 13–117 minutes latency, and in 8 GB Control SAS repair regressed by rejecting 2 correct drafts and accepting wrong revisions; (iii) *Source-symbolic quotient deduplication* offered zero rescuable candidates within actual verification budgets (0/24 rescuable, p = 1.0000); (iv) *Offline table-header materialization* recovered 4/73 fitting cases (p = 0.1250, non-significant), all 183 added quantity occurrences being already visible in baseline source spans.

4. Validated Significant Baseline Improvements: Statistically significant gains were achieved through standard supervised fine-tuning and prompt engineering: 3B Drafter QLoRA adaptation on Dev24 (0/24 → 11/24, p = 0.00098), 3B Binary Critic SFT on controlled probe (12/24 → 23/24, p = 0.00098) and natural24 transfer (3/24 → 11/24 correct deliveries, p = 0.0078), 128-token analysis prompting on 14B SAS (+17.50 pp, $p = 1.94\times10^{-6}$) and 7B SAS (+25.00 pp, p = 0.0313), and external MortarBench evaluation (SAS 39/90 vs MAS 26/90, p = 0.0192).

16 GB PRIMARY DELIVERED CORRECT

SAS 78 vs MAS 5

Δ = −45.63 pp • Exact $p = 2.1\times10^{-22}$
Report-Cluster 95% CI: [−53.90, −36.55]

16 GB PRIMARY DELIVERED WRONG

SAS 48 vs MAS 3

Δ = −28.12 pp • Exact $p = 6.8\times10^{-13}$
MAS withholds 152/160 (95.00%)

MULTI-STAGE REPAIR (BOTH TIERS)

$\Delta_{correct}$ = 0.00%

Exact p = 1.0000 in all runs
Adds +16% to +64% token overhead

STANDARD SFT ADAPTATION (N=24)

$\Delta_{accuracy}$ = +45.83%

0/24 → 11/24 • Exact p = 0.00098
χ^2 = 11.00 (p = 0.00091)

1. Mathematical & Statistical Testing Standards

For paired binary outcomes on N matched evaluation cases with 2×2 transition matrix [[a, b], [c, d]] (where b denotes discordant gains 0→1 and c denotes discordant losses 1→0 with $n_d = b + c$), and for 3×3 trinomial delivery states (*Correct*, *Wrong*, *Withheld*) with transition counts n_{ij}:

1. Exact McNemar (Paired Binomial) Test (H_0: π = 0.5):

$$p_{\text{exact, 2-sided}} = \min(1,\ 2\ \Sigma_{k=0..\min(b,c)}\ C(n_d,\ k)\ 0.5^{n_d})$$

2. Asymptotic & Edwards Continuity-Corrected χ^2 (df = 1):

$$\chi^2 = (b - c)^2 / (b + c), \quad \chi^2_{cc} = (|b - c| - 1)^2 / (b + c)$$

3. Paired Risk Difference (Δ) & Wald 95% CI:

$$\hat{\Delta} = (b - c)/N, \quad SE(\Delta) = N^{-1}\ \sqrt{[\ (b + c) - (b - c)^2/N\ ]}$$

4. Bowker Symmetry & Stuart–Maxwell Marginal Homogeneity (3×3):

$$Q_{Bowker} = \Sigma_{i<j}\ (n_{ij} - n_{ji})^2 / (n_{ij} + n_{ji}) \sim \chi^2_{df=3}$$

$$Q_{SM} = \mathbf{d}^T\ \mathbf{V}^{-1}\ \mathbf{d} \sim \chi^2_{df=2} \quad (d_i = n_{i\cdot} - n_{\cdot i},\ V_{ii} = n_{i\cdot} + n_{\cdot i} - 2n_{ii})$$

5. Cluster-Robust Whole-Report Bootstrap 95% CI:

Resampling G report clusters with replacement (10,000 draws, seed 42).

2. 16 GB Tier Evaluation (N = 160 Questions, 50 Reports, 36 Companies, 960 Runs)

The 16 GB deployment evaluation evaluated SAS (pretrained Qwen2.5-14B-Instruct Q4_K_M) against resident MAS (pretrained Qwen2.5-1.5B F16 extractor + 3B F16 drafter + 0.5B F16 verifier) across 160 public FinQA development-source questions from 50 whole company-year reports under a **11.50 GB process-tree RSS cap** (CPU 8 threads, 4096 context, 2048 token ceiling, `use_mmap=False`). Six operating profiles were evaluated (960 total case runs):

Profile & Architecture Arm	Reported Metric (C / W / H)	Raw Execution Metric (C / W / H)	Coverage (C+W)/N	Precision Delivered	Validity q / Correct p	Accept a (Corr)	Accept b (Wrong)	Prompt / Gen Tokens	Inference Time (h)	Peak RSS (GB)
SAS Primary (128-tok analysis + 2 demos)	**78 / 48 / 34**	38 / 88 / 34	78.75%	61.90%	0.938 / 0.560	0.929	0.727	642k / 32.7k	11.55 h	10.062 GB
MAS Primary (128-tok analysis + 2 demos)	**5 / 3 / 152**	5 / 3 / 152	5.00%	62.50%	0.062 / 0.500	1.000	0.600	747k / 29.5k	2.43 h	10.929 GB
SAS Direct Control (No analysis, no demos)	50 / 66 / 44	14 / 102 / 44	72.50%	43.10%	0.825 / 0.394	0.962	0.825	296k / 14.4k	5.27 h	10.040 GB
MAS Direct Control (No analysis, no demos)	0 / 5 / 155	0 / 5 / 155	3.13%	0.00%	0.031 / 0.000	—	1.000	391k / 8.4k	0.82 h	10.901 GB
SAS Repair Ablation (Primary + full repair)	79 / 50 / 31	38 / 91 / 31	80.62%	61.24%	0.938 / 0.567	0.929	0.769	760k / 35.2k	13.49 h	10.074 GB
MAS Repair Ablation (Primary + full repair)	5 / 3 / 152	5 / 3 / 152	5.00%	62.50%	0.050 / 0.625	1.000	1.000	1,223k / 33.9k	3.68 h	10.913 GB

Primary Paired Contrasts (MAS Minus SAS, N = 160 Questions, 50 Reports, 36 Companies)

Evaluation Endpoint (MAS – SAS)	Count Δ	Rate Δ̂ (SE)	Wald 95% CI	Report-Cluster 95% CI	Company-Cluster 95% CI	Exact McNemar p	Cluster Sign-Flip p	Holm-Adjusted p	Conclusion
Reported: Correct Delivered	**−73**	−45.625 pp (3.48 pp)	[−52.45, −38.80]	[−53.90, −36.55]	[−53.47, −36.44]	2.12×10^{-22}	1.53×10^{-5}	**6.10×10^{-5}**	SAS STRONGER (P < 0.0001)
Reported: Wrong Delivered	**−45**	−28.125 pp (3.66 pp)	[−35.31, −20.94]	[−37.09, −20.13]	[−38.36, −19.76]	6.82×10^{-13}	1.53×10^{-5}	**6.10×10^{-5}**	FEWER WRONG (WITHHOLDING)
Reported: Withheld	**+118**	+73.750 pp (3.48 pp)	[+66.93, +80.57]	[+64.71, +81.93]	[+66.03, +81.10]	6.02×10^{-36}	1.53×10^{-5}	—	ACCOUNTING IDENTITY
Raw: Correct Delivered	**−33**	−20.625 pp (3.46 pp)	[−27.41, −13.84]	[−28.26, −13.37]	[−28.15, −14.07]	2.10×10^{-9}	1.53×10^{-5}	**6.10×10^{-5}**	SAS STRONGER (P < 0.0001)
Raw: Wrong Delivered	**−85**	−53.125 pp (3.94 pp)	[−60.85, −45.40]	[−61.90, −44.08]	[−61.99, −43.08]	5.17×10^{-26}	1.53×10^{-5}	**6.10×10^{-5}**	FEWER WRONG (WITHHOLDING)

3×3 Trinomial Transition Matrices (SAS Rows → MAS Columns) & Symmetry Tests

Reported Metric Transitions (N = 160):

SAS \ MAS	Correct	Wrong	Withheld	Total
Correct	5	1	72	78
Wrong	0	2	46	48
Withheld	0	0	34	34
Total	5	3	152	160

Q_{Bowker} = **119.00** (df = 3, $p = 1.27\times10^{-25}$), Q_{SM} = **118.03** (df = 2, $p = 2.34\times10^{-26}$).

Raw Execution Metric Transitions (N = 160):

SAS \ MAS	Correct	Wrong	Withheld	Total
Correct	4	0	34	38
Wrong	1	3	84	88
Withheld	0	0	34	34
Total	5	3	152	160

Q_{Bowker} = **119.00** (df = 3, $p = 1.27\times10^{-25}$), Q_{SM} = **118.04** (df = 2, $p = 2.33\times10^{-26}$).

Candidate Production Funnel & Gate Ceiling Decomposition (Proposition 2 & 3)

- **SAS Funnel:** 160 questions → 150 executable candidates (validity rate q = 93.75%) → 84 reported-correct candidates (correctness conditional on validity p = 56.00%) → 78 correct delivered (gate accept rate a = 92.86%), 48 wrong delivered (gate false-accept rate b = 72.73%), 34 withheld (6 correct candidates, 18 wrong candidates, 10 unparseable/insufficient).
- **MAS Funnel:** 160 questions → **10 executable candidates** (validity rate q = 6.25%, with 150 cases terminating with `INSUFFICIENT` decisions from the 3B drafter) → 5 reported-correct candidates (p = 50.00%) → 5 correct delivered (a = 100.0%), 3 wrong delivered (b = 60.0%), 152 withheld (0 correct candidates, 2 wrong candidates, 150 insufficient).

- **Fixed-Candidate Gating Ceiling:** By Proposition 2, $|A \cap D| \leq |A|$. Because MAS produces only 5 correct candidates, any verification gate acting on those same candidates can deliver at most 5/160 correct answers. Stricter verification cannot remedy candidate production failure.
- **Retrieval Strata Breakdown:** In 138/160 cases where 100% of annotated gold evidence was retrieved in the top-8 spans, SAS delivers 74 correct answers; MAS delivers 5 correct answers and emits 128 `INSUFFICIENT` decisions. In 22/160 cases with incomplete retrieval, SAS delivers 4 correct and MAS delivers 0.

Prompting & Repair Ablation Analyses (16 GB)

- **Reasoned Prompt Bundle (128-tok Analysis + 2 Demos vs Direct Control):** SAS gains **+28 correct deliveries** (50 → 78, $\hat{\Delta}$ = +17.50 pp, Report CI [+9.76, +24.55], Exact McNemar $p = 1.94 \times 10^{-6}$) and reduces wrong deliveries by −18 (66 → 48, $\hat{\Delta}$ = −11.25 pp, p = 0.0039). MAS gains +5 correct deliveries (0 → 5, $\hat{\Delta}$ = +3.125 pp, Report CI [0.00, +6.98], p = 0.0625) and reduces wrong deliveries by −2 (5 → 3, p = 0.7266). Demonstrations and analysis jointly provide significant reasoning improvements.
- **Multi-Stage Program Repair (Repair Ablation vs Primary):** SAS attempts 34 repairs, yielding +1 reported correct (78 → 79, p = 1.0000) and **+2 wrong deliveries** (48 → 50, p = 0.5000), while under raw execution adding 0 correct and +3 wrong (88 → 91, p = 0.2500) at an extra cost of +18.2% prompt tokens and +1.95 h compute. MAS attempts 152 repairs, and **every single delivery state remains exactly identical** (5 C, 3 W, 152 H; $\hat{\Delta}$ = 0.0000, p = 1.0000 across all 160 cases), while valid candidate programs dropped from 10 to 8 at an extra cost of +63.8% prompt tokens and +1.25 h compute. Repair fails to achieve Pareto superiority in both architectures.

Scoring-Convention Concordance & Runtime Economics (16 GB)

- **Metric Disagreement Accounting:** Among 126 answers delivered by primary SAS, 34 are correct under both metrics, 44 are correct only under reported-unit scoring, 4 are correct only under raw execution, and 44 are wrong under both (concordance = 61.9%, discordance = 38.1%). For primary MAS, 5 are correct under both, 0 discordant, and 3 wrong under both. Proposition 8 bounds the net divergence: $|78 - 38| = 40 \leq 48$.
- **Inference Economics:** Primary SAS requires 259.8 s/case and **532.9 s per reported-correct answer**. Primary MAS requires 54.7 s/case, but due to low useful yield requires **1,751.4 s per reported-correct answer** (MAS is 3.29× more expensive in compute per useful answer delivered). Primary peak resident RSS is 10.062 GB for SAS (14B Q4, 8.99 GB disk) versus 10.929 GB for MAS (1.5B/3B/0.5B F16, 10.27 GB disk).

3. 8 GB Tier Evaluation (Selection52, Reserved49, Dev24, Synthetic175)

The 8 GB research evaluation evaluated systems across two distinct natural FinQA test partitions (Selection52 and Reserved49) under a **6.40 GB process-tree RSS cap**, alongside development ablations and historical synthetic benchmarks:

Partition & Evaluated Condition (8 GB Tier)	N	Reported Metric (C / W / H)	Raw Execution Metric (C / W / H)	Coverage (C+W)/N	Precision Delivered	Prompt / Gen Tokens	Inference Time (s)	Peak RSS (GB)
Selection52: Baseline Joint SAS (7B Q4, direct)	52	8 / 25 / 19	10 / 23 / 19	63.46%	24.24%	134k / 3.7k	3,439.7 s	5.348 GB
Selection52: Control SAS (Matched 64-up continuation)	52	8 / 23 / 21	10 / 21 / 21	59.62%	25.81%	144k / 4.0k	3,742.2 s	5.356 GB
Selection52: Crossed SAS (Crossed counterfactual negs)	52	**10 / 16 / 26**	12 / 14 / 26	50.00%	38.46%	152k / 4.3k	3,902.9 s	5.368 GB
Selection52: Baseline SAS (128 analysis + 2 demos)	52	9 / 24 / 19	10 / 23 / 19	63.46%	27.27%	282k / 10.5k	7,532.4 s	5.368 GB
Selection52: Control SAS (128 analysis + 2 demos)	52	9 / 20 / 23	11 / 18 / 23	55.77%	31.03%	311k / 11.1k	8,380.0 s	5.365 GB
Selection52: Crossed SAS (128 analysis + 2 demos)	52	9 / 10 / 33	10 / 9 / 33	36.54%	47.37%	333k / 11.4k	8,853.9 s	5.370 GB
Selection52: Control MAS (Matched 48-up critic)	52	3 / 19 / 30	5 / 17 / 30	42.31%	13.64%	163k / 4.7k	2,050.7 s	5.060 GB
Selection52: Crossed MAS (Crossed counterfactual critic)	52	3 / 15 / 34	5 / 13 / 34	34.62%	16.67%	170k / 4.9k	2,136.1 s	5.067 GB
Selection52: Strong Original Critic MAS (393-up baseline)	52	4 / 22 / 26	6 / 20 / 26	50.00%	15.38%	154k / 4.5k	1,929.1 s	5.056 GB
Reserved49: Control SAS (Direct drafting)	49	8 / 20 / 21	14 / 14 / 21	57.14%	28.57%	151k / 4.2k	3,890.3 s	5.368 GB
Reserved49: Crossed SAS (Direct drafting)	49	9 / 20 / 20	15 / 14 / 20	59.18%	31.03%	153k / 4.2k	3,931.8 s	5.364 GB
Reserved49: Control MAS (Direct drafting)	49	4 / 17 / 28	8 / 13 / 28	42.86%	19.05%	166k / 4.3k	2,027.4 s	5.052 GB
Reserved49: Crossed MAS (Direct drafting)	49	4 / 9 / 36	7 / 6 / 36	26.53%	30.77%	179k / 4.9k	2,222.9 s	5.064 GB

Statistical Assessment of Key 8 GB Paired Comparisons

Comparison (Right – Left) & Partition	N	Delivered Correct $\hat{\Delta}(b, c)$	Exact McNemar p	Delivered Wrong $\hat{\Delta}(b, c)$	Exact McNemar p	Report-Cluster 95% CI	Statistical Decision
Crossed vs Control MAS (Selection52)	52	0.00 pp (0, 0)	1.0000	−7.69 pp (1, 5)	0.2188	[−17.39 pp, +2.04 pp]	NOT SIGNIFICANT (P = 0.22)
Crossed vs Strong Original MAS (Selection52)	52	−1.92 pp (0, 1)	1.0000	−13.46 pp (3, 10)	0.0923	[−25.49 pp, −1.85 pp]	FAILS NO-LOSS GATE ($\Delta_{CORR} < 0$)
Crossed vs Control SAS (Selection52, Direct)	52	+3.85 pp (2, 0)	0.5000	−13.46 pp (4, 11)	0.1185	[−26.79 pp, 0.00 pp]	NOT SIGNIFICANT (P = 0.12)
Crossed vs Baseline SAS (Selection52, Direct)	52	+3.85 pp (2, 0)	0.5000	−17.31 pp (4, 13)	**0.0490**	[−29.09 pp, −5.36 pp]	SIG WRONG REDUCTION VS BASE
Crossed vs Control SAS (Selection52, Reasoned)	52	0.00 pp (1, 1)	1.0000	−19.23 pp (3, 13)	**0.0213**	[−38.03 pp, −2.08 pp]	SIG WRONG REDUCTION (REASONED)
Crossed vs Control SAS (Reserved49, Direct)	49	+2.04 pp (1, 0)	1.0000	0.00 pp (3, 3)	1.0000	[−10.64 pp, +8.62 pp]	EXACT ZERO ERROR DIFFERENCE
Crossed vs Control MAS (Reserved49, Direct)	49	0.00 pp (0, 0)	1.0000	−16.33 pp (2, 10)	**0.0386**	[−26.23 pp, −4.26 pp]	WITHHOLDS MORE; RAW DROPS −1
Control MAS vs Control SAS (Selection52 Arch)	52	−9.62 pp (0, 5)	0.0625	−7.69 pp (4, 8)	0.3877	[−18.33 pp, −2.00 pp]	SAS LEADS CORRECT (P = 0.063)

Comparison (Right – Left) & Partition	N	Delivered Correct Δ̂ (b, c)	Exact McNemar *p*	Delivered Wrong Δ̂ (b, c)	Exact McNemar *p*	Report-Cluster 95% CI	Statistical Decision
Crossed MAS vs Crossed SAS (Selection52 Arch)	52	−13.46 pp (0, 7)	**0.0156**	−1.92 pp (6, 7)	1.0000	[−28.85 pp, −2.00 pp]	SAS DELIVERS SIG MORE CORRECT
Crossed MAS vs Crossed SAS (Reserved49 Arch)	49	−10.20 pp (1, 6)	0.1250	−22.45 pp (3, 14)	**0.0127**	[−38.71 pp, −7.50 pp]	MAS WITHHOLDS 73.5% (P = 0.0009)

Additional 8 GB Diagnostic & Adaptation Evaluations

- **3B Drafter QLoRA Fine-Tuning (Dev24):** Unadapted 3B drafter (all abstained, 0/24) versus QLoRA-adapted 3B drafter (11/24 correct candidates; 10 delivered correct, 11 wrong, 3 withheld). Discordant gains $b = 11$, $c = 0$ ($\hat{\Delta} = +45.83$ pp, Exact McNemar $p =$ **0.00098**, $\chi^2 = 11.00$). Confirms standard parameter adaptation is highly effective under memory constraints.
- **Trained 3B Binary Critic Probes (Dev24):** On balanced controlled single-edit mutation probes (12 report pairs, 24 programs), trained critic achieved 11/12 pair separations and 23/24 gate classifications versus 0/12 and 12/24 for unadapted critic ($p =$ **0.00098**). Transferred to natural drafter errors (Natural24), correct deliveries rose from 3/24 to 11/24 ($p =$ **0.0078**), but wrong deliveries also increased from 0/24 to 4/24 (Bowker $Q = 12.00$, $p = 0.0025$; break-even error penalty $\lambda = 2.0$).
- **Question-Blind Shortcut Audit (N = 382):** Depth-3 decision trees using only program structure features classified 72.0% of original synthetic validation mutations ($Z = 8.60$, $p = 8.28 \times 10^{-18}$) and 77.2% of extended mutations ($Z = 10.64$, $p = 1.90 \times 10^{-26}$). Construction of balanced crossed quartets successfully neutralized this shortcut to exactly 50.0% ($Z = 0.00$). On fixed 32-review probes, SAS 7B switched verdicts with question in 15/16 cases ($p = 0.00052$) and Original MAS 3B in 14/16 ($p = 0.00418$), demonstrating that neural critics attend to financial questions rather than relying solely on program syntax.
- **Shared-Backbone 3B Role Serving:** Serial switching between actor and critic adapters on a single shared 3B backbone produced exact zero logit divergence (max $|\Delta \text{ logit}| = 0.00000$) and identical 32-token continuations while reducing peak resident tree RSS from 4.831 GB to 2.571 GB (−2.260 GB, −46.78%).
- **Historical Synthetic Benchmark (N = 175) & External Baselines:** On MortarBench (N = 90), SAS achieved 39/90 (43.3%) exact matches versus MAS 26/90 (28.9%) ($\hat{\Delta} = -14.44$ pp, Exact McNemar $p =$ **0.0192**). On FRAMES closed-book negative control (N = 100), SAS scored 2/100 and MAS 1/100 ($p = 1.0000$).

4. Cross-Tier Comparative Synthesis & Comparability Analysis

Project Agere evaluated systems across two distinct hardware deployment environments: nominal 8 GB (simulated at 6.40 GB RSS cap) and nominal 16 GB (evaluated at 11.50 GB RSS cap). The table below details the exact configuration inventory across both tiers:

Dimension / Attribute	Historical 8 GB Benchmark	Redeveloped 8 GB Natural Study	Primary 16 GB Deployment Study
Target Task & Data Source	Synthetic KYC onboarding & SME credit memos	FinQA Public Training-Source Reports	FinQA Public Development-Source Reports
Unique Evaluation Questions	175 synthetic cases (100 KYC + 75 Credit)	101 questions (52 selection + 49 reserved)	160 questions (50 whole reports, 36 companies)
Cohort Overlap with 16 GB	0 overlapping tasks	0 overlapping questions, 0 overlapping reports	Mutually disjoint from 8 GB development partitions
Single-Agent Architecture (SAS)	Qwen2.5-7B-Instruct (GGUF Q4_K_M)	Qwen2.5-7B-Instruct (Q4_K_M + trained LoRA)	Qwen2.5-14B-Instruct (GGUF Q4_K_M, pretrained)
Multi-Agent Architecture (MAS)	1.5B (Drafter) + 0.5B (Extr) + 0.5B (Verif) F16	Shared 3B Q4 role backbone + LoRA adapters	1.5B (Extr) + 3B (Drafter) + 0.5B (Verif) distinct F16
Role Model Residency	All 3 models resident simultaneously	Single resident 3B base; serial adapter swap	All 3 specialist models resident simultaneously
Context Window / Generation Ceiling	2048 tokens / 2048 generated tokens	4096 tokens / 2048 generated tokens	4096 tokens / 2048 generated tokens
Process-Tree RSS Ceiling	6.40 GB (8.0 GB physical workstation parity)	6.40 GB (monitored parent + worker tree)	11.50 GB (15.44 GB physical host parity)
Model Adaptation Status	Unadapted base checkpoints	Task-adapted QLoRA drafters & critics	Pretrained base checkpoints (both arms)
Inference Runtime & Engine	Windows 11 llama.cpp backend	Ubuntu Linux native AVX2 backend	Ubuntu Linux portable AVX2 CPU backend
Causal RAM Scaling Identified?	**No** (different tasks and metrics)	**No** (train-source questions, adapted models)	**No** (dev-source questions, pretrained models)

> **Scientific Non-Identification of Causal RAM-Scaling Laws:** Because model parameter scale (7B vs 14B SAS; 3B vs 5B MAS), precision formats (Q4 vs F16 MAS), training/adaptation status (adapted vs pretrained), evidence retrieval prompts (compact catalog vs suggestion stage), and question cohorts (train-source vs dev-source) vary simultaneously between the 8 GB and 16 GB evaluations, the cross-tier differences **cannot be parameterized into a causal RAM-scaling response curve**. Attempting to fit a cross-tier regression on these data conflates hardware capacity with model architecture, precision, and task distribution shifts.

Resident Process-Tree RSS vs. Disk Storage Footprint

A critical engineering finding across both tiers is that nominal parameter counts and GGUF file sizes on disk do not reliably predict resident RAM requirements when precision formats differ:

- **16 GB Tier Memory Footprint:** 14B Q4 SAS has a model file size of **8.99 GB** on disk and reaches a peak resident process-tree RSS of **10.062 GB**. In contrast, MAS uses three smaller models totaling 5.0B parameters, but because they are resident in unquantized FP16, their disk footprint is **10.27 GB** and their peak resident RSS reaches **10.929 GB**. Thus, the multi-agent team consumes **+867 MB more resident RAM** than the 14-billion-parameter monolith.
- **8 GB Tier Memory Footprint:** SAS 7B Q4 consumes 4.68 GB on disk and reaches 5.348–5.370 GB peak RSS. When MAS loads separate resident specialist instances, it reaches 5.05–5.54 GB RSS. However, when implementing the shared-backbone role-switching backend, peak RSS drops to **2.571 GB**, freeing over 2.26 GB of memory headroom while maintaining bit-level logit parity.

5. Formal Resource-Constrained Selective-Answering Framework

To analyze the trade-offs between answer accuracy, error exposure, and abstention under hardware constraints, we formulate eight conditional mathematical propositions (proved and evaluated in `research/le16_v2/THEORETICAL_FRAMEWORK.md`):

Proposition 1: Exact Endpoint Accounting & Rejection Confounding.
For any fixed evaluation cohort of N questions and any evaluation metric, every case occupies exactly one mutually exclusive delivery state: Delivered Correct (C), Delivered Wrong (W), or Withheld (H).

$$C + W + H = N \implies \Delta(C/N) + \Delta(W/N) + \Delta(H/N) = 0$$

Consequence: A reduction in wrong deliveries ($\Delta W < 0$) can always be trivially manufactured by increasing withholding ($\Delta H > 0$) without any improvement in semantic correctness. Therefore, wrong-delivery reductions cannot be claimed as an accuracy improvement unless accompanied by non-decreasing correct deliveries ($\Delta C \geq 0$).

Proposition 2: Fixed-Candidate Gating Ceiling.
Let A denote the set of correct candidate programs generated by an actor, and let D denote the set of candidates delivered by a verification gate that can only accept or reject candidates without modifying their text.

$$C_{gate} = |A \cap D| \leq |A|$$

Consequence: On 16 GB Primary MAS, the actor produces only $|A| = 5$ correct candidates out of 160 questions. Even an oracle verifier can deliver at most 5/160 correct answers (3.13%). The observed low delivery rate is mathematically bounded by candidate production, not gate selectivity.

Proposition 3: Candidate-Production and Acceptance Chain-Rule Decomposition.
Let *q* be the fraction of questions yielding an executable candidate, *p* be correctness conditional on execution validity, *a* be gate acceptance among correct candidates, and *b* be false-acceptance among wrong candidates:

$$C/N = q \cdot p \cdot a, \qquad W/N = q \cdot (1 - p) \cdot b, \qquad H/N = 1 - C/N - W/N$$

Empirical 16 GB Primary Values: SAS: q = 0.9375, p = 0.5600, a = 0.9286, b = 0.7273 &implies; C/N = 78/160 (48.75%), W/N = 48/160 (30.00%).
MAS: q = 0.0625, p = 0.5000, a = 1.0000, b = 0.6000 &implies; C/N = 5/160 (3.13%), W/N = 3/160 (1.88%).

Proposition 4: Linear Utility Switch Points as Operational Preference Curves.
Let benefit of correct delivery be normalized to 1, penalty for wrong delivery be $\lambda \geq 0$, and cost of withholding be 0: $U(\lambda) = (C - \lambda W)/N$.

$$U_{SAS}(\lambda) - U_{MAS}(\lambda) = [\ (C_{SAS} - C_{MAS}) - \lambda(W_{SAS} - W_{MAS})\ ] / N \ \ \&implies; \ \ \lambda^* = (C_{SAS} - C_{MAS}) / (W_{SAS} - W_{MAS})$$

16 GB Primary Switch Points: Reported Metric: $\lambda^* = (78 - 5)/(48 - 3) =$ **73/45 ≈ 1.622**. Raw Metric: $\lambda^* = (38 - 5)/(88 - 3) =$ **33/85 ≈ 0.388**.
If withholding carries human review cost $\kappa \geq 0$, $U(\lambda, \kappa) = (C - \lambda W - \kappa H)/N$, shifting the switch point to $\lambda^*(\kappa) = [73 + 118\kappa] / 45$. Choosing λ post-hoc is preference fitting, not a discovered mathematical safety boundary.

Proposition 5: Repair Break-Even and Empirical Sample Dominance.
For repair to achieve a utility gain over primary no-repair with marginal compute cost $c_{compute} \geq 0$: $\Delta U_{repair}(\lambda) = (\Delta C - \lambda \Delta W)/N - c_{compute} > 0$.
16 GB SAS: Reported: ΔC = +1, ΔW = +2 &implies; $\Delta U = (1 - 2\lambda)/160 - c < 0$ for all $\lambda > 0.5$. Raw: ΔC = 0, ΔW = +3 &implies; $\Delta U = -3\lambda/160 - c \leq 0$ for all $\lambda \geq 0$.
16 GB MAS: ΔC = 0, ΔW = 0, ΔH = 0 &implies; $\Delta U = -c_{compute} < 0$. Primary no-repair empirically dominates repair ablation across all positive compute costs.

Proposition 6: Memory Storage Lower Bounds for Distinct Weight Tensors.
For *k* distinct resident models with tensor parameters n_j stored at effective bits per element b_j: $M_{total} \geq \Sigma_j (n_j\ b_j / 8) + M_{KV} + M_{buffers}$.
Consequence: Resident FP16 MAS (1.5B + 3B + 0.5B = 5.0B params × 16 bits = 10.0 GB weights) requires strictly more memory than 14B Q4 SAS (14B params × 4.5 bits ≈ 7.88 GB weights), confirming why MAS resident RSS exceeds SAS despite fewer nominal parameters.

Proposition 7: Bounded Independent-Cluster Hoeffding Uncertainty.
Let report-level contrast contributions $D_g \in [-1, 1]$ with report size weights $w_g = n_g/N$ ($\Sigma\ w_g = 1$). By Hoeffding's inequality:

$$Pr(|S - E[S]| \geq \varepsilon) \leq 2\ exp(-\varepsilon^2\ /\ [2\ \Sigma_g\ w_g^2]), \qquad \varepsilon_\alpha = \sqrt{[\ 2\ \Sigma_g\ w_g^2 \cdot ln(2/\alpha)\ ]}, \qquad G_{effective} = 1\ /\ \Sigma_g\ w_g^2$$

Evaluation: On 16 GB (50 reports), $G_{eff} = 36.99$ &implies; conservative 95% bounded half-width $\varepsilon_{0.05} =$ **44.66 pp**. For company clustering (36 companies), $G_{eff} = 19.88$ &implies; $\varepsilon_{0.05} =$ **60.98 pp**.

Proposition 8: Metric Disagreement Accounting.
Let R_{only} denote deliveries correct only under reported scoring and E_{only} correct only under raw execution:

$$C_{reported} - C_{raw} = R_{only} - E_{only} \ \ \&implies; \ \ |C_{reported} - C_{raw}| \leq R_{only} + E_{only}$$

Evaluation: On 16 GB Primary SAS, $R_{only} = 44$, $E_{only} = 4$ &implies; 78 − 38 = 40 ≤ 48.

6. Master Research Hypothesis Ledger

The table below provides a comprehensive synthesis of every major research hypothesis, candidate method, and engineering mechanism evaluated across the entire research program:

Research Hypothesis / Mechanism	Evaluated Scope	Effect Size (Δ)	Statistical Test & p-Value	Status Category	Scientific & Practical Conclusion
1. MAS Superiority over SAS at Parity Memory (Resident FP16 MAS vs Quantized SAS)	16 GB (N = 160) 8 GB (N = 52, 49)	Δ_{corr} = −45.63 pp (16G) Δ_{corr} = −9.62 to −13.46 pp (8G)	16G: p = 2.12×10^{-22} (Holm $p < 0.0001$) 8G: p = 0.0156 to 0.0625	REJECTED (SAS DOMINATES)	Quantized SAS achieves substantially higher useful correct coverage across both tiers. MAS withholds 73%–95% of questions.
2. Multi-Stage Iterative Program Repair (Verifier-guided program revision)	16 GB (N = 160) 8 GB (N = 52, 24)	Δ_{corr} = 0.00 pp (All) Δ_{wrong} = +1.25 to +3.85 pp	Exact McNemar p = 1.0000 SAS Wrong: p = 0.25 to 0.50	CONFIRMED NULL / NEGATIVE	Repair produced zero accuracy improvement in any condition, added 16%–64% token cost, and caused correct-to-wrong regressions in SAS.
3. Crossed-Question Verifier Curriculum (Counterfactual negative critic training)	8 GB (N = 52, 49) SAS & MAS arms	MAS: Δ_{corr} = 0 to −1.92 pp SAS: Δ_{wrong} = −13.46 to −19.23 pp	MAS vs Orig: p = 1.0000 ($\Delta_{corr} < 0$) SAS vs Base: p = 0.0013 to 0.0490	ERROR TRADE-OFF / NOT PARETO	Reduces wrong deliveries in SAS but failed no-loss adoption gate in MAS (lost 1 correct answer) and did not replicate error drops on reserved set.
4. Bounded Source-Quotient Deduplication (Symbolic program equivalence repair pools)	8 GB (N = 24) 4 grouping rules	0 / 24 rescuable within actual budget	Oracle Bound p = 1.0000 Top-4 reps changed in 1/24 cases	ZERO BUDGET HEADROOM	No initially wrong case has a numerically correct candidate in any of the four-check / 8192-token representative allocations.
5. Offline Header / Catalog Materialization (First-row preservation & span-value catalogs)	8 GB (N = 73) Fitting partition	Mappable: 41 → 45 / 73 (Δ = +5.48 pp)	Exact McNemar p = 0.1250 Sign test tokens: p = 0.1686	NON-SIGNIFICANT / PRE-NEURAL	Not statistically significant; 100% of added quantities already visible in baseline spans; no neural QA improvement evaluated.
6. Standard Parameter Fine-Tuning (QLoRA) (3B drafter & critic SFT)	8 GB (N = 24) Dev24 paired	Drafter: 0 → 11 / 24 (+45.8 pp) Critic: 12 → 23 / 24 (+45.8 pp)	Exact McNemar p = 0.00098 χ^2 = 11.00 (p = 0.00091)	CONFIRMED SIGNIFICANT (P < 0.001)	Task-specific parameter adaptation provides large, statistically robust gains under memory caps.
7. Structured Analysis Prompting Bundle (128-tok analysis + 2 demos vs direct)	16 GB (N = 160) 8 GB (N = 24)	16G SAS: +17.50 pp correct 8G SAS: +25.00 pp correct	16G: p = 1.94×10^{-6} (Holm < 0.001) 8G: p = 0.0313	CONFIRMED SIGNIFICANT (P < 0.001)	Intermediate program reasoning with demonstrations significantly boosts useful answer generation in single-agent architectures.
8. Shared-Backbone Role-Switching Serving (Single resident base + serial adapter swap)	8 GB deployment 2 prompts, 4 switches	Δ RSS = −2.26 GB (−46.8%) max |Δ logit| = 0.00000	Exact Deterministic Identity (13,660+ memory samples)	VERIFIED ENGINEERING PARITY	Eliminates redundant base weight duplication for multi-role workflows without logit drift or KV cross-contamination.
9. External Task Generalization (MortarBench & FRAMES baselines)	Mortar (N = 90) FRAMES (N = 100)	Mortar: SAS 39 vs MAS 26 FRAMES: SAS 2 vs MAS 1	Mortar: Exact p = 0.0192 FRAMES: Exact p = 1.0000	SAS LEADS EXTERNAL TASK	MAS suffered severe retrieval/schema failures on MortarBench (emitted empty lists on 57/57 transaction questions).
10. Cross-Tier RAM-Scaling Law (Parametric scaling curves across tiers)	Cross-Tier (8 GB vs 16 GB)	Confounded parameter sets (scale, precision, prompt, split)	Non-Identified Hypothesis (No valid pooled regression)	SCIENTIFICALLY NON-IDENTIFIED	Because model size, precision, adaptation status, and question cohorts differ simultaneously across tiers, causal RAM effects cannot be isolated.